

ASSESSMENT OF ROLE OF METHYL COBALAMINE IN EXCITOTOXIC EFFECTS OF GLUTAMICACID

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ABSTRACT:

Neurotoxicants and environmental pollutants cause neuronal damage and may damage myelin sheath. Since Methyl cobalamine (B_{12}) has effect on myelin sheath formation it was hypothesized that B_{12} administration in case of nerve damage by some neurotoxicants may be protected or corrected by B_{12} administration. (Braun, 1977 and Braun, 1980). For generations physicians have recommended B-vitamins especially B_{12} for neuropsychiatric symptoms. In present study neuroprotective role of B_{12} is assessed. Present work was planned to evaluate the therapeutic or prophylactic role of B_{12} in nerve damage by neurotoxicant i.e. glutamic acid. Present study revealed no significant neuroprotection or defensive regimen offered by B_{12} for neurotoxic damage. Glutamic acid is probably affecting neurons or dentritis sparing myelin sheath. Neuroprotection expected by B_{12} may be applied only to myelin sheath.

INTRODUCTION

Neurotoxicology is an exciting area of science not only because of the importance of toxic injury to the nervous system in human disease, but also because of specific toxins have been invaluable tools for the advancement of neurobiology. In fact, much of our understanding of organization and function of the nervous system is based on observations derived from the action of neurotoxins. The binding of exogenous compounds to membranes has been the basis for the definition of specific receptors within the brain; an understanding of the roles of different cell types in the function of the nervous system has stemmed from the selectivity of certain toxicants in injuring only certain cell types; and important differences in basic metabolic requirements of different sub populations of neurons have been inferred from effects of toxicants (Braun, 1977).

Neurotoxic exposures of humans are a major environmental concern. Human exposures have occurred throughout the world in epidemics in which thousands of individuals have been exposed to neurotoxicants; and

perhaps of even greater concern, large numbers of people develop neurologic disorders and are exposed to environments containing known neurotoxicants and many more compounds whose effects on the nervous system are unknown. The exact cause of Parkinson's disease is unknown, but epidemiologic data suggest that the incidence is related to exposure to neurotoxins such as Methyl phenyltetra pyridinium (MPTP) and glutamate (Becking *et al.*, 1993).

MATERIALS AND METHODS

Experimental Animals:

Locally bred 25 Albino-Wistar rats were used for the acute and repeated Methycobalamine administration respectively. Female and male rats (purchased from HEJ Research Institute of Chemistry) weighing from 200-300Gms. The rats were caged individually in specifically designed cages with saw-dust covered floor, in a quiet room with access to water and cubes of standard rat food at room temperature. All animals were handled 2 days before starting the experiment, so that, rats could adapt to the novel environment. Rats were divided into three groups as follow:

GROUP 1:

Control rats were given normal diet daily. Water intake was not measured. The body weight of rats was monitored daily and their growth rate was calculated. Catalepsy (frozen body posture) was assessed after 30 minutes of dose for 3 minutes. Behavioral changes and locomotor activity monitored daily.

GROUP 2:

Tested rats were given Glutamic Acid at a dose of 0.2 ml of 5 mmol/litre once daily. Rats received dose between 8:00-9:00 a.m. Water intake was not measured. The body weight of rats was monitored daily and their growth rate was calculated. Catalepsy was assessed after 30 minutes of doses for 3 minutes. Behavioral changes and locomotor activity monitored daily after 45 minutes

GROUP 3:

Tested rats were given Glutamic Acid orally at a dose of 0.2 ml of 5 mmol/litre and Methylcobalamine at a dose of 6 μ g/300gm intraperitoneally once daily. Rats received dose

between 8:00-9:00 a.m. Water intake was not measured. The body weight of rats was monitored daily and their growth rate was calculated. Catalepsy was measured by positive geotropism after 30 minutes of dose for 3 minutes. Behavioral changes and locomotor activity monitored daily after 45 minutes of injection.

Drugs and Experimental Dose:

Drugs used for the treatment were:

1. Methylcobolamin (Macter) available in 500 μ g/ml ampoules was injected. The drug was prepared in water for injection.
2. Glutamic Acid (Purchased from Sigma) was used. The drug was prepared in deionized water and administered orally at a dose of 0.2 ml of 5 mmol/litre.

Adaptation Phase:

Rats were kept in Perspex transparent cages (26x26x26) after injection for 15 minutes 10 get them familiarized with environment. These cages were used as home cages.

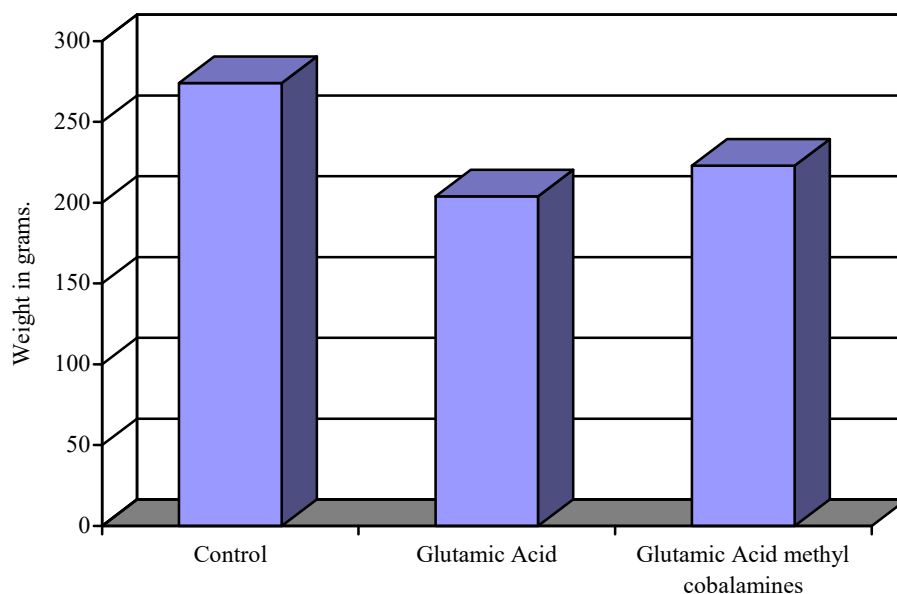


Fig. 1: Effect of excitotoxic amino acid and methyl cobalamines on weight of rats. Values are $\pm$ S.D. (n=6). Significant differences by student t-test $P < 0.05$ considered significant as compared to respective control.

EXPERIMENTAL PROTOCOL

Chronic Dose of Glutamic Acid:

Rats were randomly assigned as control and test. Control rats were given normal diet only and test rats were given Glutamic Acid orally between 9.00 a.m. – 10.00 a.m. Catalepsy was monitored 30 minutes post dose. Cut off time 3 minutes employed. After 15 minutes animals were injected with methylcobalamine at a dose of 6µg. Animals were observed after 45 minutes for activity measurement (Open field and cage crossing and gross behavior).

Chronic Dose of Glutamic Acid and Methycobolamin:

25 rats were randomly assigned to group 1 and group 2 and group 3. First two I groups contained 10 animals and 3rd group consist of 5 rats (i.e. controlled group). Rats in controlled group received nothing but normal diet, 2nd group received 2 weeks treatment of Glutamic Acid orally and 3rd group were received 2 weeks treatment with Glutamic Acid orally and Methylcobalamine.

1) Measurement of Body Weight:

Rats were weighed before starting the

experiment. The body weight in both groups was monitored daily. Growth rate was calculated as percentage of the Average day's body weight.

2) Catalepsy:

The typical effect of administering a dopaminergic receptor antagonist is a suppression of spontaneous exploratory and locomotion behavior and the excitation of a state known as catalepsy (Mason, 1984). Catalepsy in rats is generally regarded as a valid model for detecting the EPS liability of compounds in humans (Drinkenburg, 1999). Catalepsy refers to a marked difficulty in initiating voluntary motor activity, usually demonstrated experimentally by showing that the subject does not change position when placed in an awkward, presumably uncomfortable posture. Nevertheless, the subject is not asleep, and in fact can be aroused to move by an appropriate stimulus.

3) Inclined Screen Test

Animals were placed on an inclined surface. Inclination was about 60° in such a way that animal face upward.

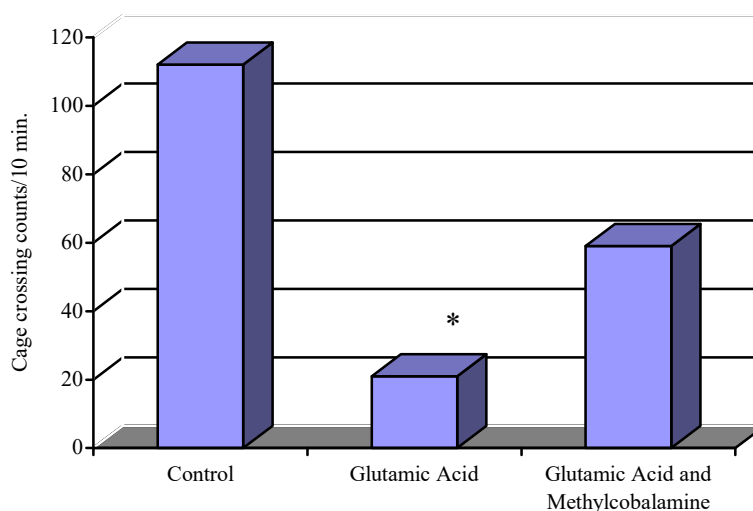


Fig. 2: Effect of excitotoxic amino acid and methyl cobalamines on cage crossing. Values are $\pm$ S.D. (n=6) statistically significant * $P < 0.05$ from respective control animals.

4) Rightening Reflex:

Animals were placed gently on their back on an iron surface. Loss of rightening reflex was to have occurred if animal remained on its back for more than 30 seconds. Sleeping time was measured as the time elapsed between the loss and regaining of rightening reflex.

5) Traction Test:

The test material was examined for their muscle strength. The treated animals as well as controls were suspended by means of their forepaws to a metallic wire stretched horizontally across the cage. The numbers of animals (after administration of drug) which were incapable of touching the wire with at least one of the hind paw were noted.

RESULTS AND DISCUSSION

Glutamate neurotransmission is mediated by at least three subtypes of receptors that may

be recognized by their specificity for kainite, quisqualate, and NMDA (Lipton and Rosenberg, 1994). The entry of glutamate in to the central nervous system (CNS) is regulated at the blood brain barrier, and following an injection of a large dose of glutamate exerts its effects in the area of the brain in which the blood brain barrier is least developed, the circumventricular organ (Farrel and Risau, 1994). Within the site of limited access, glutamate injures neurons, apparently by opening glutamate dependent ion channels, ultimately leading to neuronal swelling and neuronal cell death (Coyle, 1987). The toxicity affects of the dendrites and neuronal cell bodies but seems to spare axons. The only none related human condition is the "Chinese Restaurant Syndrome" in which consumption of large amounts of mono sodium glutamate (MSG) as a seasoning may lead to burning sensation in the face, neck, and chest (Choi, 1988).

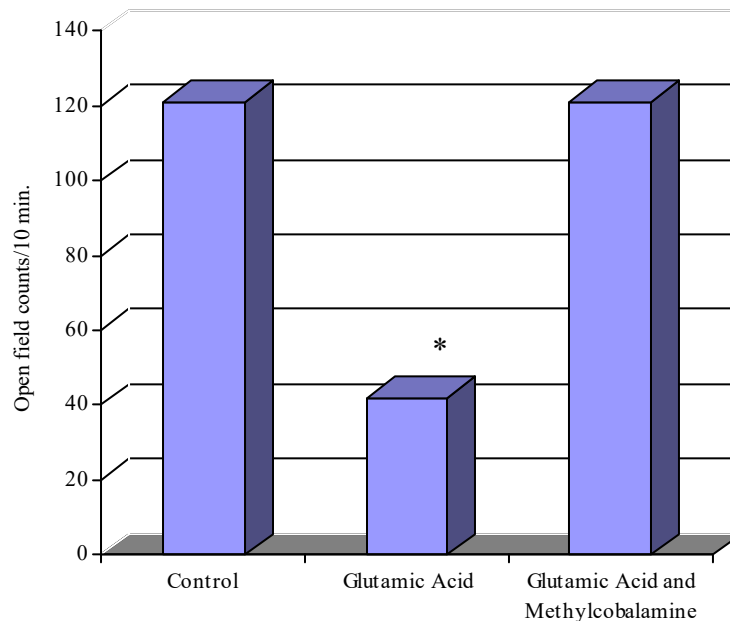


Fig. 3: Effect of excitotoxic amino acid and methyl cobalamines on open field activity. (Values are $\pm$ S.D. (n=6) significant difference by student t-test * $P < 0.05$ from respective control animals)

The cyclic glutamate analog kainite was initially isolated from seaweed in Japan as the active component of this herbal treatment of ascariasis. Kainite is extremely potent as an excitotoxin, being 100 fold more toxic than glutamate and is selective at a molecular level for the kainite receptor (Coyle, 1987). Like glutamate, kainite selectively injures dendrites and neurons, and shows no substantial effect on glia or axons. As a result, this compound has found great use in neurobiology as a tool (Brightman, 1989). Injected in to a region of the brain, kainite can destroy the neurons of that area with out disrupting all of the fibers that pass through the same region (Coyle, 1987). Neurobiologists, as a result of this neurotoxic tool, are able to study the role of neurons in a particular area independent of the axonal injuries that occur when similar lesioning experiments are performed by mechanical cutting (Choi, 1988).

In present study following changes have been noted Glutamate decreased the weight of animals but coadministration with vitamin B₁₂ did not enhance growth rate (Fig. 1).

Cage crossing was significantly lowered by glutamate. B₁₂ administration increased cage crossing non significantly (Fig. 2). Increased open field activity exhibited was significant by glutamate which was reversed by vitamin B₁₂ administration (Fig. 3). Catalepsy was not induced by glutamate. Inclined screen test did not reveal any significant findings. Traction test did not show any significant change. Righting reflex was not present even after glutamate administration and coadministration with vitamin B₁₂.

Present work was designed to evaluate the damages caused by chronic administration of glutamic acid in rats. It was also planned to observe the effects of cobalamines (viamin B12) on toxic effect induced by glutamic acid or on general and behavioral changes. Vitamin B₁₂ administration was expected to have some beneficial effects through increasing pre-cursors of nucleic acid and myelin sheath formation. Probably the damage was caused

only to neurons or dentritis and myelin sheath was probably not damaged (Meadows *et al.*, 1994). Anabolic action of B₁₂ on myelin sheath fail to show any protection in damage caused by calcium overload induced by glutamate (Braun *et al.*, 1980 and Duncan *et al.*, 1987).

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